Hybrid Fractional-Order PSO–Grey Wolf Optimization of FOPID Controllers for Trajectory Tracking of a 6-DOF Robotic Arm

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Abstract. A robotic arm is a multi-input multi-output system with strong nonlinearity and coupling, and high-precision trajectory tracking requires carefully tuned joint controllers. This paper designs fractional-order PID (FOPID) controllers for the six joints of a 6-DOF robotic arm and proposes FOPSO-GWO to tune their 30 parameters. FOPSO-GWO is a hybrid algorithm that introduces fractional-order velocity memory into particle swarm optimization (PSO) and combines it with the hunting mechanism of the grey wolf optimizer (GWO). The fitness function evaluates the step and sinusoidal tracking performance of all joints and limits the peak driving torque and overshoot of each joint. Simulation results show that, compared with PID and FOPID control, the FOPID controller tuned by FOPSO-GWO reduces the average overshoot by 73 % and 53 %, the adjustment time by 46 % and 31 % and the mean squared error of sinusoidal tracking by 91 % and 77 %.

Keywords: 6-DOF robotic arm, trajectory tracking, FOPID controller, particle swarm optimization, grey wolf optimizer

1 Introduction

With the development of intelligent manufacturing, robotic arms are used more and more widely. A robotic arm is a complex system with multiple inputs and outputs, strong nonlinearity and strong coupling, so high-precision trajectory tracking control is a key problem in its application. PID control is widely used in industry because of its simple structure and model independence [1], but its performance is limited for strongly coupled nonlinear systems. The fractional-order PID (FOPID) controller extends the PID controller with an integral order λ and a differential order μ [12]. These two parameters give more flexibility in shaping the control performance and better robustness [9, 13, 17]. At the same time, they make parameter tuning more difficult, especially for a 6-DOF robotic arm, where the 30 parameters of six coupled joint controllers must be tuned together.

Swarm intelligence algorithms are widely used to tune FOPID controllers. Reference [18] designed a FOPID controller for a wind energy conversion system

with the grey wolf optimizer (GWO), and reference [6] tuned a FOPID controller for load frequency control with the whale optimization algorithm. GWO has also been applied to the trajectory generation of 6-DOF manipulators [2]. Particle swarm optimization (PSO) [5, 14] is simple and converges fast, but it is prone to premature convergence. Introducing fractional calculus into the velocity update gives the particles a memory of their previous velocities and improves the search [11, 3]. Combining PSO with GWO, in which the three best wolves lead the search, improves the balance between exploration and exploitation [8, 16].

This paper studies the trajectory tracking control of a 6-DOF robotic arm with FOPID controllers. The main contributions are as follows:

1. A hybrid algorithm, FOPSO-GWO, is proposed. It combines the fractional-order velocity update of PSO with the hunting mechanism of GWO.
2. A fitness function is designed that evaluates the step and sinusoidal responses of all six joints together. It also limits the peak driving torque and the overshoot of each joint.
3. Trajectory tracking simulations of the 6-DOF robotic arm verify that the FOPID controller tuned by FOPSO-GWO achieves higher control accuracy than PID and FOPID control.

2 Preliminaries

2.1 Fractional-Order Calculus

The Grünwald–Letnikov definition of the fractional derivative of order α is

$$D^{\alpha} f(t) = \lim_{h \to 0} \frac{1}{h^{\alpha}} \sum_{k=0}^{\lfloor (t-t_0)/h \rfloor} (-1)^k \binom{\alpha}{k} f(t - kh), \tag{1}$$

and a negative α gives the fractional integral. In the Laplace domain the operator is s^{α}. For implementation, s^{α} with $\alpha \in (0, 1)$ is approximated by Oustaloup's recursive filter [10]:

$$s^{\alpha} \approx \omega_h^{\alpha} \prod_{k=-N}^{N} \frac{s + \omega_k'}{s + \omega_k}, \quad \omega_k' = \omega_b \left(\tfrac{\omega_h}{\omega_b}\right)^{\frac{k+N+(1-\alpha)/2}{2N+1}}, \quad \omega_k = \omega_b \left(\tfrac{\omega_h}{\omega_b}\right)^{\frac{k+N+(1+\alpha)/2}{2N+1}}, \tag{2}$$

with $N = 5$ and the frequency band $[\omega_b, \omega_h] = [10^{-3}, 10^{3}]\,\mathrm{rad/s}$.

2.2 PID Controller and FOPID Controller

The FOPID controller of joint j acts on the tracking error $e_j = q_{r,j} - q_j$:

$$\tau_j(t) = K_{p,j}\, e_j(t) + K_{i,j}\, D^{-\lambda_j} e_j(t) + K_{d,j}\, D^{\mu_j} e_j(t), \qquad j = 1, \ldots, 6, \tag{3}$$

where K_p, K_i and K_d are the proportional, integral and differential gains. The integer-order PID controller is the special case $\lambda = \mu = 1$. The controllers run at 1 kHz. Orders above one are split into an integer part and a fractional part, which is realised by (2).

2.3 Dynamic Model of the 6-DOF Robotic Arm

The robotic arm studied is a UR5-type 6-DOF arm, with the parameters and test conditions of [4] (Table 1). Its dynamic model is [15]

$$M(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} + G(q) = \tau, \tag{4}$$

where $q \in \mathbb{R}^6$ is the vector of joint angles, M the inertia matrix, $C\dot{q}$ the Coriolis and centrifugal torques, G the gravity torques and τ the joint driving torques. The model is computed by the recursive Newton–Euler algorithm [7] and integrated with a fourth-order Runge–Kutta method with a step of 1 ms. As in the test conditions, gravity and friction are not considered.

Table 1. Parameters of the robotic arm (standard DH parameters, link masses and principal inertias)

Joint i	d_i (m)	a_i (m)	α_i (rad)	m_i (kg)	$(I_{xx}, I_{yy}, I_{zz})_i$ (kg m^2)
1	0.128	0	$\pi/2$	2.0	(0.001, 0.001, 0.001)
2	0	−0.612	0	2.5	(0.040, 0.040, 0.040)
3	0	−0.571	0	5.7	(0.060, 0.060, 0.040)
4	0.164	0	$\pi/2$	3.9	(0.055, 0.055, 0.040)
5	0.115	0	$-\pi/2$	2.5	(0.040, 0.040, 0.040)
6	0.092	0	0	2.5	(0.040, 0.040, 0.040)

3 FOPSO-GWO Algorithm

3.1 Fractional-Order PSO

In PSO, particle i moves by

$$v_i^{k+1} = w_k v_i^k + c_1 r_1 \odot (p_i - x_i^k) + c_2 r_2 \odot (g - x_i^k), \qquad x_i^{k+1} = x_i^k + v_i^{k+1}, \tag{5}$$

where p_i is the personal best of the particle, g the global best, r_1, r_2 uniform random vectors, $\odot$ the element-wise product and w_k the inertia weight, which decreases linearly from 0.9 to 0.4 [14]. Writing the velocity change as a fractional derivative of order a and truncating (1) after four terms gives the fractional-order velocity [11]:

$$w_k v_i^k \;\rightarrow\; (w_k - 1 + a)\, v_i^k + \tfrac{a(1-a)}{2}\, v_i^{k-1} + \tfrac{a(1-a)(2-a)}{6}\, v_i^{k-2} + \tfrac{a(1-a)(2-a)(3-a)}{24}\, v_i^{k-3}. \tag{6}$$

Each particle thus remembers its last four velocities.

3.2 Hybrid with the Grey Wolf Hunting Mechanism

In GWO [8], the three best wolves α, β and δ lead the pack. FOPSO-GWO takes them as the three best personal bests and builds, for each particle, the estimate

$$\hat{x}_i = \frac{1}{3} \sum_{L \in \{\alpha,\beta,\delta\}} \left(L - A_L \odot |C_L \odot L - x_i^k| \right), \qquad A_L = 2a_g r_L - a_g, \quad C_L = 2r'_L, \tag{7}$$

where $a_g = 2(1-k/K)^2$ decreases from 2 to 0 over the K iterations. The estimate is added to the velocity as a third attraction:

$$v_i^{k+1} = (6) + c_1 r_1 \odot (p_i - x_i^k) + c_2 r_2 \odot (g - x_i^k) + c_3 r_3 \odot (\hat{x}_i - x_i^k). \tag{8}$$

In early iterations $|A_L| > 1$, and the wolf term moves particles past the leaders, which favours exploration. As a_g decreases, the particles gather around the three leaders, which favours exploitation. The coefficients $c_1 = c_2 = 1.5$ and $c_3 = 1$ keep the total attraction equal to that of standard PSO ($c_1 + c_2 = 4$), and the fractional order is fixed at $a = 0.9$, which keeps the velocity memory over the whole search.

3.3 Parameter Tuning of the FOPID Controllers

Each particle encodes the 30 parameters $(K_p, K_i, K_d, \lambda, \mu)$ of the six joints, scaled to $[0, 1]$. The gains are scaled logarithmically, with $\log_{10} K_p \in [-2, 5]$, $\log_{10} K_i \in [-4, 5]$ and $\log_{10} K_d \in [-1, 3]$, and the orders linearly, with $\lambda, \mu \in [0.05, 1.95]$.

Every candidate is simulated under both the step and the sinusoidal input. Its performance indices are normalised by reference values, with the ratios ρ_i for the ITAE of both responses (Eq. (10)), the overshoot, the adjustment time, the peak time and the mean squared error. The ratios ϵ_k do the same for the absolute torque sum and the torque variation of both responses. The fitness function is

$$J = \frac{\sum_{i=1}^{6} w_i \max(\rho_i, 0.5) + \sum_{k=1}^{4} \epsilon_k}{\sum_{i=1}^{6} w_i + 4} + 2 \sum_i \max(0, \rho_i - 1) + 10\,P, \tag{9}$$

with weights $w = (0.5, 0.5, 1, 1, 1, 1)$. The second term penalises the overshoot, adjustment time, peak time, MSE and torque sum when they do not improve on their reference values. P penalises violations of the per-joint limits: the peak driving torque of each joint and an overshoot of at most 33.1 % on every joint. Unstable or numerically ill-conditioned candidates are given a large fitness value. The procedure is summarised in Algorithm 1, and the parameter settings are listed in Table 2.

Algorithm 1 FOPSO-GWO for FOPID parameter tuning

initialise N particles in $[0,1]^{30}$ around an initial FOPID controller and at random
evaluate the fitness J of every particle; set p_i and g
for $k = 1, \ldots, K$ **do**
    update w_k and a_g; select the leaders α, β, δ
    for each particle i **do**
        compute $\hat{x}_i$ by (7) and v_i^{k+1} by (8); limit $|v_i^{k+1}| \le 0.2$
        $x_i \leftarrow x_i + v_i^{k+1}$, kept inside $[0,1]^{30}$
        simulate the step and sinusoidal responses and evaluate $J(x_i)$; update p_i
    end for
    update g
end for
return the FOPID parameters encoded by g

Table 2. Parameter settings of FOPSO-GWO

Population size, maximum iterations	$N = 30$, $K = 100$
Learning factors	$c_1 = c_2 = 1.5$, $c_3 = 1$
Fractional order	$a = 0.9$
Inertia weight	$0.9 \to 0.4$, linearly decreasing
Grey wolf coefficient	$a_g = 2(1 - k/K)^2$
Maximum velocity	0.2 of each parameter range

4 Simulation Experiment of the Robotic Arm

4.1 Experimental Setup

Two inputs are applied to all six joints, each simulated for 5 s from rest:

- a step input of 1 rad at $t = 1$ s;
- a sinusoidal input $q_{r,j}(t) = \sin(1.5\,t)$ rad.

For the step response, the overshoot, the adjustment time (5 % band) and the peak time are evaluated. For the sinusoidal response, the mean squared error (MSE) of the tracking error and the sum of the absolute driving torque are evaluated. All indices are averaged over the six joints, and the integral of the time-weighted absolute error is

$$\mathrm{ITAE} = \int_0^5 t \sum_{j=1}^{6} |e_j(t)|\, dt. \tag{10}$$

Three control schemes are compared: PID control, FOPID control and the FOPID control tuned by FOPSO-GWO, denoted FOPSO-GWO-FOPID. The PID and FOPID controllers are set to the performance reported for the same arm in [4]. FOPSO-GWO was run eight times independently, and the best result is reported.

4.2 Convergence of FOPSO-GWO

Figure 1 shows the convergence curves of the fitness value. In the best run, the fitness decreases from 35.4 for the initial FOPID controller to 1.42 within 3030 fitness evaluations. All eight runs converge stably, to values between 1.42 and 2.16. The tuned parameters are listed in Table 3.

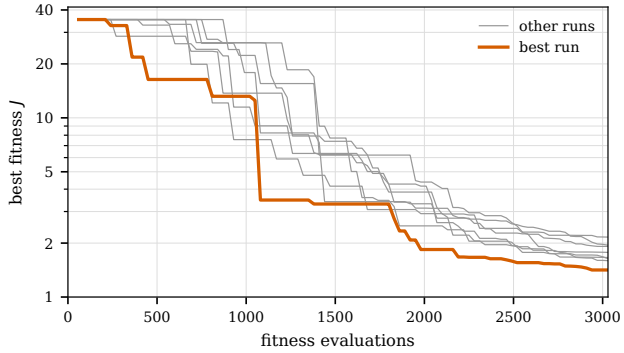


Fig. 1. Convergence curves of FOPSO-GWO (eight independent runs)

Table 3. FOPID parameters tuned by FOPSO-GWO

	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6
K_p	110.7	0.2054	0.03346	590.0	33.96	3393
K_i	42.81	132.5	0.7265	43.96	4.420	4.329×10^{-4}
K_d	156.1	989.1	890.0	9.623	171.5	0.3650
λ	1.534	0.07579	1.377	0.07649	0.5251	0.05009
μ	0.8239	0.5172	0.5289	0.9728	0.7322	1.637

4.3 Step Response

Figure 2 shows the step responses of the six joints, and Table 4 lists the average performance indices. With PID control, the overshoot of half of the joints exceeds 30 %, reaching 150.2 % on joint 5. With FOPSO-GWO-FOPID, the overshoot of every joint stays below 26 %, at 14.3 %, 17.6 %, 0.1 %, 25.7 %, 6.7 % and 17.3 %. On average, the overshoot is reduced by 73.3 % and 53.4 %, the adjustment time by 45.8 % and 30.6 %, and the peak time by 23.5 % and 17.8 %, compared with PID and FOPID control. The response is faster and closer to the reference trajectory. The ITAE of the step response is reduced from 4.34 (PID) and 2.16

(FOPID) to 1.19. The peak driving torque at the step instant is reduced from about 2.2×10^5 N m to 4.6×10^4 N m.

Table 4. Average performance indices of the three controllers with step response

Controller	Overshoot (%)	Adjustment time (s)	Peak time (s)	ITAE
PID	51.1	2.49	1.45	4.34
FOPID	29.2	1.95	1.35	2.16
FOPSO-GWO-FOPID	**13.6**	**1.35**	**1.11**	**1.19**

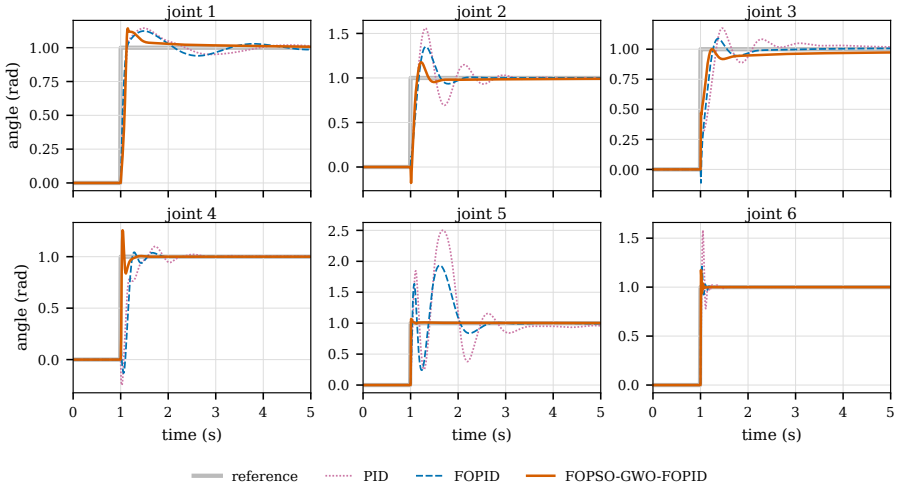


Fig. 2. Position tracking of joints 1–6 with step response

4.4 Sinusoidal Response

Figure 3 shows the tracking errors of the six joints under the sinusoidal input, and Table 5 lists the tracking performance indices. The MSE of FOPSO-GWO-FOPID is 1.88×10^{-3} rad^2, which is 91.2 % lower than that of PID control and 77.4 % lower than that of FOPID control. The ITAE is reduced from 5.81 (PID) and 4.74 (FOPID) to 1.72. At the same time, the sum of the absolute driving torque is reduced to 9.61×10^3 N m. FOPSO-GWO-FOPID therefore tracks the sinusoidal trajectory more accurately with a smaller control input.

Table 5. Tracking performance indices of the three controllers with sine response

Controller	MSE (rad^2)	Sum of $\lvert\tau\rvert$ (N m)	ITAE
PID	2.12×10^{-2}	3.50×10^{4}	5.81
FOPID	8.29×10^{-3}	2.41×10^{4}	4.74
FOPSO-GWO-FOPID	$\mathbf{1.88 \times 10^{-3}}$	$\mathbf{9.61 \times 10^{3}}$	**1.72**

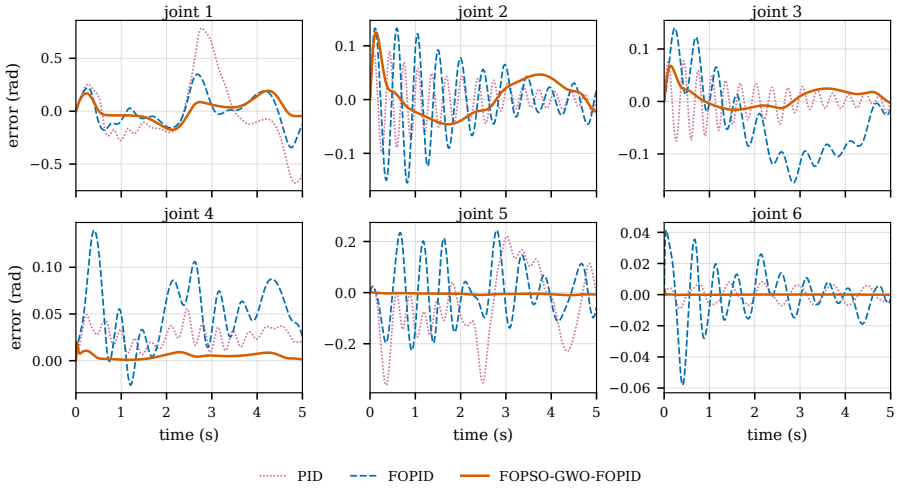


Fig. 3. Tracking error of joints 1–6 with sine response

5 Conclusion

This paper studied the trajectory tracking control of a 6-DOF robotic arm with FOPID controllers and proposed FOPSO-GWO, a hybrid of fractional-order PSO and the grey wolf optimizer, to tune their 30 parameters. The fractional-order velocity keeps a memory of the particles' previous motion, and the grey wolf mechanism guides the search with the three best solutions. The fitness function evaluates the step and sinusoidal responses of all joints and limits the driving torque and overshoot of each joint. Simulation results show that, compared with PID and FOPID control, the FOPID controller tuned by FOPSO-GWO has smaller overshoot, shorter adjustment and peak times and a smaller tracking error, which meets the high-precision control requirements of the 6-DOF robotic arm. Future work will verify the method on a physical robotic arm.

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